

```
print("Name: Gargi Sukhatankar")
print("SAPID: 60019220125")
print("Roll No: B027")
print("Branch: CSE(ICB)")
```

```
➦ Name: Gargi Sukhatankar
  SAPID: 60019220125
  Roll No: B027
  Branch: CSE(ICB)
```

```
import nltk
nltk.download('gutenberg')
nltk.download('punkt')
nltk.corpus.gutenberg.fileids()
```

```
➦ [nltk_data] Downloading package gutenberg to /root/nltk_data...
[nltk_data] Unzipping corpora/gutenberg.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
['austen-emma.txt',
 'austen-persuasion.txt',
 'austen-sense.txt',
 'bible-kjv.txt',
 'blake-poems.txt',
 'bryant-stories.txt',
 'burgess-busterbrown.txt',
 'carroll-alice.txt',
 'chesterton-ball.txt',
 'chesterton-brown.txt',
 'chesterton-thursday.txt',
 'edgeworth-parents.txt',
 'melville-moby_dick.txt',
 'milton-paradise.txt',
 'shakespeare-caesar.txt',
 'shakespeare-hamlet.txt',
 'shakespeare-macbeth.txt',
 'whitman-leaves.txt']
```

```
from nltk.corpus import gutenberg
caesar = gutenberg.words('shakespeare-caesar.txt')
print(caesar[:20])
```

```
➦ ['I', 'The', 'Tragedie', 'of', 'Julius', 'Caesar', 'by', 'William', 'Shakespeare', '1599', ']', 'Actus', 'Primus', '.', 'Scoena', 'f
```

```
import nltk
```

```
# Download the necessary resources
nltk.download('gutenberg')
nltk.download('punkt')
nltk.download('punkt_tab') # Download punkt_tab data for sentence tokenization

# nltk.corpus.gutenberg.fileids()

# from nltk.corpus import gutenberg
# caesar = gutenberg.words('shakespeare-caesar.txt')
# print(caesar[:20])
```

```
➦ ['Hence', ':', 'home', 'you', 'idle', 'Creatures', ',', 'get', 'you', 'home', ':', 'Is', 'this', 'a', 'Holiday', '?']
[nltk_data] Downloading package gutenberg to /root/nltk_data...
[nltk_data] Package gutenberg is already up-to-date!
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
```

```
from nltk import bigrams
first_sentence = gutenberg.sents('shakespeare-caesar.txt')[5]
print(first_sentence)
```

```
➦ ['Hence', ':', 'home', 'you', 'idle', 'Creatures', ',', 'get', 'you', 'home', ':', 'Is', 'this', 'a', 'Holiday', '?']
```

```
from collections import defaultdict
print(list(bigrams(first_sentence, pad_left=True, pad_right=True)))
```

```
➦ [(None, 'Hence'), ('Hence', ':'), (':', 'home'), ('home', 'you'), ('you', 'idle'), ('idle', 'Creatures'), ('Creatures', ','), (',',
```

```
model = defaultdict(lambda: defaultdict(lambda: 0))
for sentence in gutenbergsents('shakespeare-caesar.txt'):
    words = [word.lower() for word in sentence if word.isalpha()]

    for w1, w2 in bigrams(words, pad_left=True, pad_right=True):
        model[w1][w2] += 1
print(model["you"]["idle"])
print(model[None]["the"])
```

→ 1
22

```
for w1 in model:
    total_w1 = float(sum(model[w1].values()))
    for w2 in model[w1]:
        model[w1][w2] /= total_w1

print(model["you"]["idle"])
print(model[None]["the"])
```

→ 0.0025575447570332483
0.010171058714748035